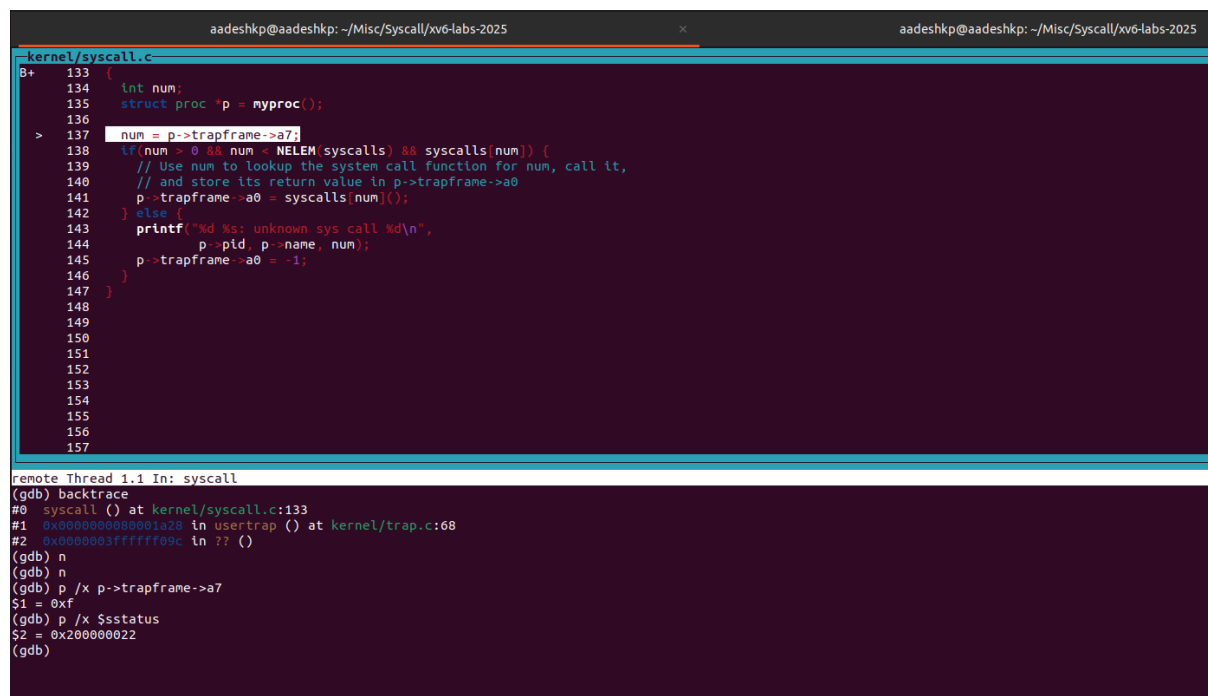


Make Grade

```
make[1]: Leaving directory '/home/aadeshkp/Misc/Syscall/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (4.7s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.7s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (0.7s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.6s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.0s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.0s)
== Test time ==
time: OK
Score: 45/45
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025$
```

Exercise 1

Troubleshooting in GDB



The screenshot shows a terminal window with two panes. The top pane displays the source code for `kernel/syscall.c`, with line 137 highlighted: `num = p->trapframe->a7;`. The bottom pane shows a GDB session for the `remote Thread 1.1 In: syscall`. The GDB commands and output are as follows:

```
(gdb) backtrace
#0 syscall () at kernel/syscall.c:133
#1 0x00000000000001a28 in usertrap () at kernel/trap.c:68
#2 0x00000003ffffff09c in ?? ()
(gdb) n
(gdb) n
(gdb) p /x p->trapframe->a7
$1 = 0xf
(gdb) p /x $sstatus
$2 = 0x200000022
(gdb)
```

Modifying `syscall.c` to cause kernel to panic.

```
GNU nano 6.2 syscall.c
[SYS_pipe] sys_pipe,
[SYS_read] sys_read,
[SYS_kill] sys_kill,
[SYS_exec] sys_exec,
[SYS_fstat] sys_fstat,
[SYS_chdir] sys_chdir,
[SYS_dup] sys_dup,
[SYS_getpid] sys_getpid,
[SYS_sbrk] sys_sbrk,
[SYS_pause] sys_pause,
[SYS_uptime] sys_uptime,
[SYS_open] sys_open,
[SYS_write] sys_write,
[SYS_mknod] sys_mknod,
[SYS_unlink] sys_unlink,
[SYS_link] sys_link,
[SYS_mkdir] sys_mkdir,
[SYS_close] sys_close,
};

void
syscall(void)
{
    int num;
    struct proc *p = myproc();

    num = p->trapframe->a7;
    if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
        // Use num to lookup the system call function for num, call it,
        // and store its return value in p->trapframe->a0
        p->trapframe->a0 = syscalls[num]();
    } else {
        printf("kd %s: unknown sys call %d\n",
            p->pid, p->name, num);
        p->trapframe->a0 = -1;
    }
}
```

```
rcpny -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pi
e -no-pie -c -o user/forphan.o user/forphan.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_forphan user/forphan.o user/lib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_forphan > user/forphan.asm
riscv64-unknown-elf-objdump -t user/forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/forphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -ncmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-st
rcpny -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pi
e -no-pie -c -o user/dorphan.o user/dorphan.c
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/lib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -ncmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-st
rcpny -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pi
e -no-pie -c -o user/attack.o user/attack.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_attack user/attack.o user/lib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_attack > user/attack.asm
riscv64-unknown-elf-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/attack.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -ncmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-st
rcpny -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pi
e -no-pie -c -o user/secret.o user/secret.c
riscv64-unknown-elf-objdump -S user/_secret > user/secret.asm
riscv64-unknown-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_int user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_userte
sts user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_attack user/_secret
meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 972 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -devi
ce virtio-blk-device,drives=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
scause=0xd sepc=0x80001c7a stval=0x0
panic: kerneltrap
```

Checking which process was running during panic

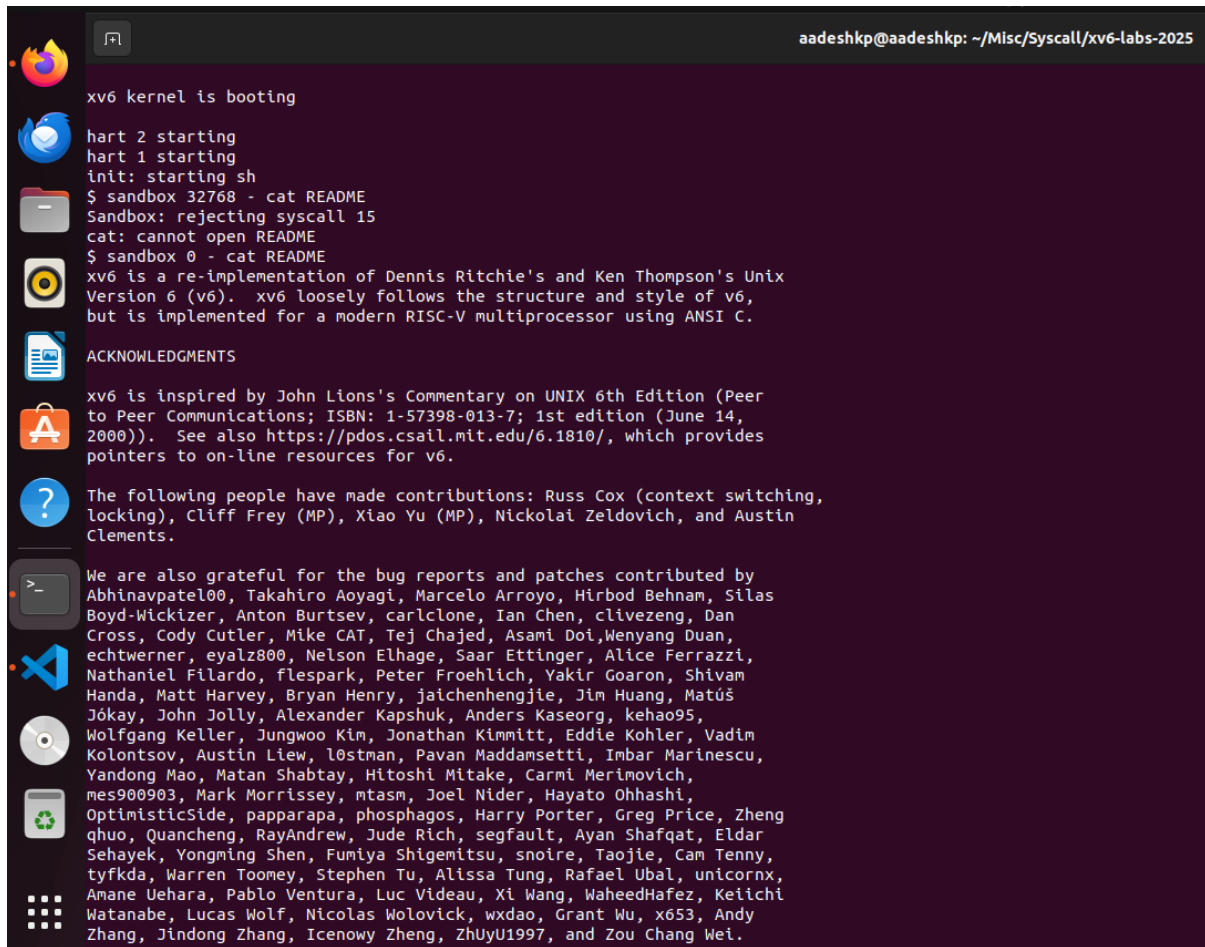
```
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025/kernel
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025

This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from kernel/kernel...
warning: File "/home/aadeshkp/Misc/Syscall/xv6-labs-2025/.gdbinit" auto-loading has been declined by your 'auto-load safe-path' set to "$debugdir:$datadir/auto-load".
To enable execution of this file add
  add-auto-load-safe-path /home/aadeshkp/Misc/Syscall/xv6-labs-2025/.gdbinit
line to your configuration file "/home/aadeshkp/.config/gdb/gdbinit".
To completely disable this security protection add
  set auto-load safe-path /
line to your configuration file "/home/aadeshkp/.config/gdb/gdbinit".
For more information about this security protection see the
"Auto-loading safe path" section in the GDB manual.  E.g., run from the shell:
  info "(gdb)Auto-loading safe path"
(gdb) target remote localhost:26000
Remote debugging using localhost:26000
0x80000137 in ?? ()
(gdb) b 0x80001c7a
Function "0x80001c7a" not defined.
Make breakpoint pending on future shared library load? (y or [n]) n
(gdb) b *0x80001c7a
Breakpoint 1 at 0x80001c7a: file kernel/syscall.c, line 137.
(gdb) c
Continuing.
[Switching to Thread 1.2]

Thread 2 hit Breakpoint 1, syscall () at kernel/syscall.c:137
137      num = *(int *)0;
(gdb) p p->name
$1 = "init", '\000' <repeats 11 times>
(gdb) p p->pid
$2 = 1
(gdb)
```

Exercise 2



The image shows a terminal window with a dark purple background. On the left side, there is a vertical column of application icons: a Firefox icon, a Telegram icon, a file manager icon, a music player icon, an Apple App Store icon, a help icon, a terminal icon, a VS Code icon, a CD icon, and a recycling icon. At the top right of the terminal, the text 'aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025' is displayed. The terminal output consists of several lines of text: 'xv6 kernel is booting', 'hart 2 starting', 'hart 1 starting', 'init: starting sh', '\$ sandbox 32768 - cat README', 'Sandbox: rejecting syscall 15', 'cat: cannot open README', '\$ sandbox 0 - cat README', 'xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix Version 6 (v6). xv6 loosely follows the structure and style of v6, but is implemented for a modern RISC-V multiprocessor using ANSI C.', 'ACKNOWLEDGMENTS', 'xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer to Peer Communications; ISBN: 1-57398-013-7; 1st edition (June 14, 2000)). See also <https://pdos.csail.mit.edu/6.1810/>, which provides pointers to on-line resources for v6.', 'The following people have made contributions: Russ Cox (context switching, locking), Cliff Frey (MP), Xiao Yu (MP), Nickolai Zeldovich, and Austin Clements.', and a long list of names under the heading 'We are also grateful for the bug reports and patches contributed by'. The list of names includes: Abhinavpatel00, Takahiro Aoyagi, Marcelo Arroyo, Hirbod Behnam, Silas Boyd-Wickizer, Anton Burtsev, carlclone, Ian Chen, clivezeng, Dan Cross, Cody Cutler, Mike CAT, Tej Chajed, Asami Doi, Wenyang Duan, echtwerner, eyalz800, Nelson Elhage, Saar Ettinger, Alice Ferrazzi, Nathaniel Filardo, flespark, Peter Froehlich, Yakir Goaron, Shivam Handa, Matt Harvey, Bryan Henry, jaichenhengjie, Jim Huang, Matúš Jókay, John Jolly, Alexander Kapshuk, Anders Kaseorg, kehao95, Wolfgang Keller, Jungwoo Kim, Jonathan Kimmitt, Eddie Kohler, Vadim Kolontsov, Austin Liew, l0stman, Pavan Maddamsetti, Imbar Marinescu, Yandong Mao, Matan Shabtay, Hitoshi Mitake, Carmi Merimovich, mes900903, Mark Morrissey, mtasm, Joel Nider, Hayato Ohhashi, OptimisticSide, papparapa, phosphagos, Harry Porter, Greg Price, Zheng qhuo, Quancheng, RayAndrew, Jude Rich, segfault, Ayan Shafqat, Eldar Sehayek, Yongming Shen, Fumiya Shigemitsu, snoire, Taojie, Cam Tenny, tyfkda, Warren Toomey, Stephen Tu, Alissa Tung, Rafael Ubal, unicornx, Amane Uehara, Pablo Ventura, Luc Videau, Xi Wang, WaheedHafez, Keiichi Watanabe, Lucas Wolf, Nicolas Wolovick, wxdao, Grant Wu, x653, Andy Zhang, Jindong Zhang, Icenowzy Zheng, ZhuyU1997, and Zou Chang Wei.

```
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 - cat README
Sandbox: rejecting syscall 15
cat: cannot open README
$ sandbox 0 - cat README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
but is implemented for a modern RISC-V multiprocessor using ANSI C.

ACKNOWLEDGMENTS

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to Peer Communications; ISBN: 1-57398-013-7; 1st edition (June 14,
2000)). See also https://pdos.csail.mit.edu/6.1810/, which provides
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The following people have made contributions: Russ Cox (context switching,
locking), Cliff Frey (MP), Xiao Yu (MP), Nickolai Zeldovich, and Austin
Clements.

We are also grateful for the bug reports and patches contributed by
Abhinavpatel00, Takahiro Aoyagi, Marcelo Arroyo, Hirbod Behnam, Silas
Boyd-Wickizer, Anton Burtsev, carlclone, Ian Chen, clivezeng, Dan
Cross, Cody Cutler, Mike CAT, Tej Chajed, Asami Doi, Wenyang Duan,
echtwner, eyalz800, Nelson Elhage, Saar Ettinger, Alice Ferrazzi,
Nathaniel Filardo, flespark, Peter Froehlich, Yakir Goaron, Shivam
Handa, Matt Harvey, Bryan Henry, jaichenhengjie, Jim Huang, Matúš
Jókay, John Jolly, Alexander Kapshuk, Anders Kaseorg, kehao95,
Wolfgang Keller, Jungwoo Kim, Jonathan Kimmitt, Eddie Kohler, Vadim
Kolontsov, Austin Liew, l0stman, Pavan Maddamsetti, Imbar Marinescu,
Yandong Mao, Matan Shabtay, Hitoshi Mitake, Carmi Merimovich,
mes900903, Mark Morrissey, mtasm, Joel Nider, Hayato Ohhashi,
OptimisticSide, papparapa, phosphagos, Harry Porter, Greg Price, Zheng
qhuo, Quancheng, RayAndrew, Jude Rich, segfault, Ayan Shafqat, Eldar
Sehayek, Yongming Shen, Fumiya Shigemitsu, snoire, Taojie, Cam Tenny,
tyfkda, Warren Toomey, Stephen Tu, Alissa Tung, Rafael Ubal, unicornx,
Amane Uehara, Pablo Ventura, Luc Videau, Xi Wang, WaheedHafez, Keiichi
Watanabe, Lucas Wolf, Nicolas Wolovick, wxdao, Grant Wu, x653, Andy
Zhang, Jindong Zhang, Icenowzy Zheng, ZhuyU1997, and Zou Chang Wei.
```

Exercise 3

```
hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 README cat README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
but is implemented for a modern RISC-V multiprocessor using ANSI C.
```

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```
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025
```

```
Yandong Mao, Matan Shabtay, Hitoshi Mitake, Carmi Merimovich,
mes900903, Mark Morrissey, mtasm, Joel Nider, Hayato Ohhashi,
OptimisticSide, papparapa, phosphagos, Harry Porter, Greg Price, Zheng
qhuo, Quancheng, RayAndrew, Jude Rich, segfault, Ayan Shafqat, Eldar
Sehayek, Yongming Shen, Fumiya Shigemitsu, snoire, Taojie, Cam Tenny,
tyfkda, Warren Toomey, Stephen Tu, Alissa Tung, Rafael Ubal, unicornx,
Amane Uehara, Pablo Ventura, Luc Videau, Xi Wang, WaheedHafez, Keiichi
Watanabe, Lucas Wolf, Nicolas Wolovick, wxdao, Grant Wu, x653, Andy
Zhang, Jindong Zhang, Icenowy Zheng, ZhUYU1997, and Zou Chang Wei.
```

ERROR REPORTS

Please send errors and suggestions to Frans Kaashoek and Robert Morris (kaashoek,rtm@mit.edu). The main purpose of xv6 is as a teaching operating system for MIT's 6.1810, so we are more interested in simplifications and clarifications than new features.

BUILDING AND RUNNING XV6

You will need a RISC-V "newlib" tool chain from <https://github.com/riscv/riscv-gnu-toolchain>, and qemu compiled for riscv64-softmmu. Once they are installed, and in your shell search path, you can run "make qemu".

```
$ sandbox 32768 README cat sh
cat: cannot open sh
$ sandbox 128 sh sh
$ sandbox 128 sh ls
usertrap(): unexpected scause 0xd pid=9
sepc=0x113c stval=0x9091300001917
$ QEMU: Terminated
```

Exercise 4

```
aadeshkp@aadeshkp: ~/Misc/Syscall/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strcmp -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/attack.o user/attack.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_attack user/attack.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_attack > user/attack.asm
riscv64-unknown-elf-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* //; /^$/d' > user/attack.syn
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_sleep user/_tressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1011 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ secret AadeshSecretPW
$ attack
0
$ attack
AadeshSecretPW
$
```